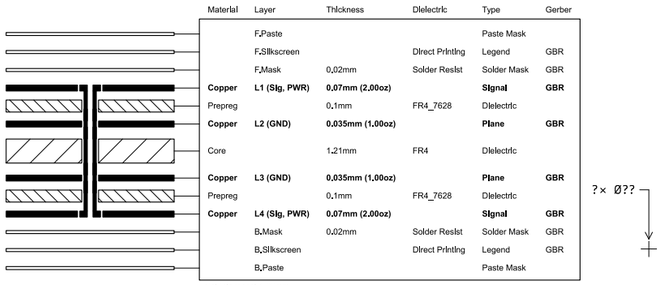
BLE/Thread/Zigbee EndDevice Fabrication Document

Layer Stack Legend

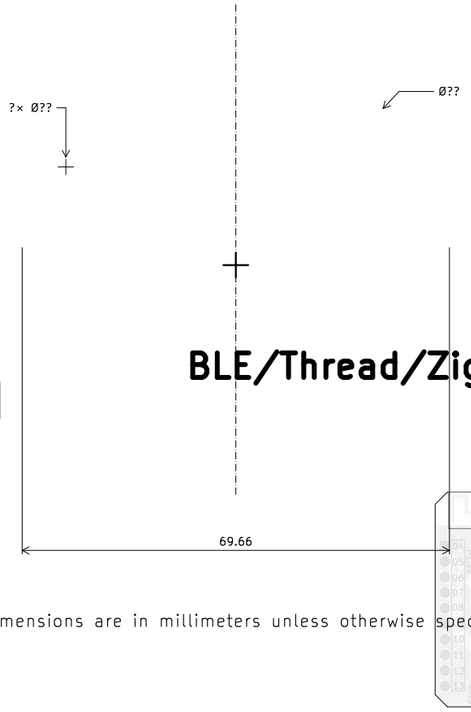


Material	Layer	Thickness	Dielectric	Type	Gerber
	F_Paste			Paste Mask	
	F_Silkscreen		Direct Printing	Legend	GBR
	F_Mask	0,02mm	Solder Reslst	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0,07mm (2,00oz)**		**Signal**	**GBR**
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	**L2 (GND)**	**0,035mm (1,00oz)**		**Plane**	**GBR**
Core		1,21mm	FR4	Dielectric	
Copper	**L3 (GND)**	**0,035mm (1,00oz)**		**Plane**	**GBR**
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	**L4 (Sig, PWR)**	**0,07mm (2,00oz)**		**Signal**	**GBR**
	B_Mask	0,02mm	Solder Reslst	Solder Mask	GBR
	B_Silkscreen		Direct Printing	Legend	GBR
	B_Paste			Paste Mask	

Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	50	±10 %	L1	0,2	0,15	L4



All dimensions are in millimeters unless otherwise specified.

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.
2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
4) SURFACE FINISH: IMMERSION GOLD
5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
9) PCB MATERIAL REQUIREMENTS:
   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY GOODBYTE.
10) DESIGN TO MEET MINIMUM CAPABILITIES:

BOARD SIZE	34.999 × 34.999 mm
BOARD THICKNESS	1.660 mm
TRACE WIDTH	0.150 mm
TRACE TO TRACE	0.100 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	0.900 mm
ANNULAR RING	0.075 mm
COPPER TO HOLE	0.200 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
12) CONFIRM SPACE WIDTHS AND SPACINGS.

Top Fabrication (Scale 1:1)

Project Name: **Wireless demo board**		Board Name: **BLE/Thread/Zigbee EndDevice**	{ GoodByte embedded software }
Revision: + (Unreleased)	Rollout Stage: PRELIMINARY	Git Hash: a122132	Company: Goodbyte
Comments:	Sheet Title: Fabrication	File Name: BTZ-EndDevice.kicad_pcb	Sheet Path:
	Designer: Marcin Flak	Reviewer:	Date: 2024-04-13
			Size: **A3**
			Sheet: **1** of **10**